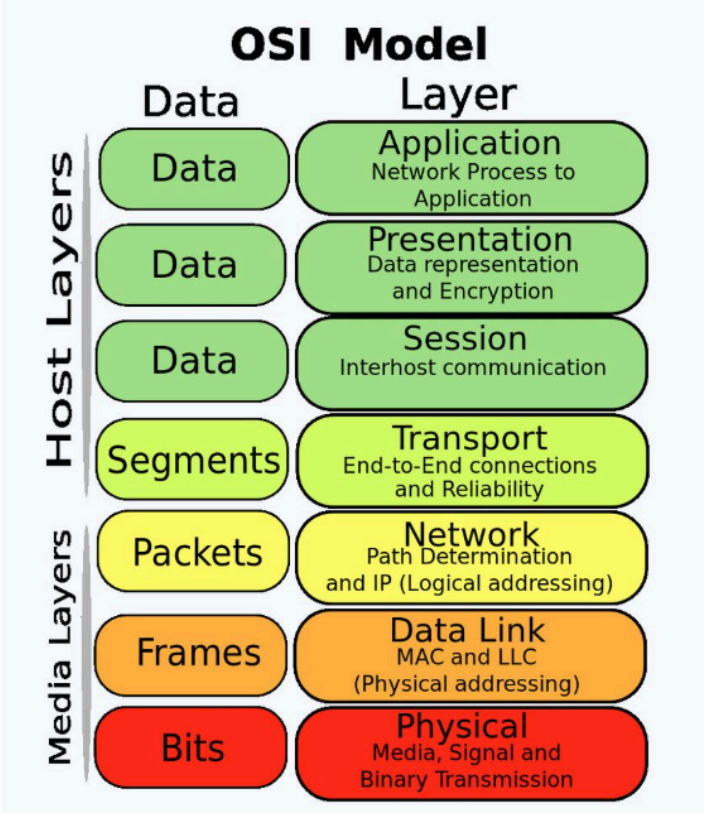
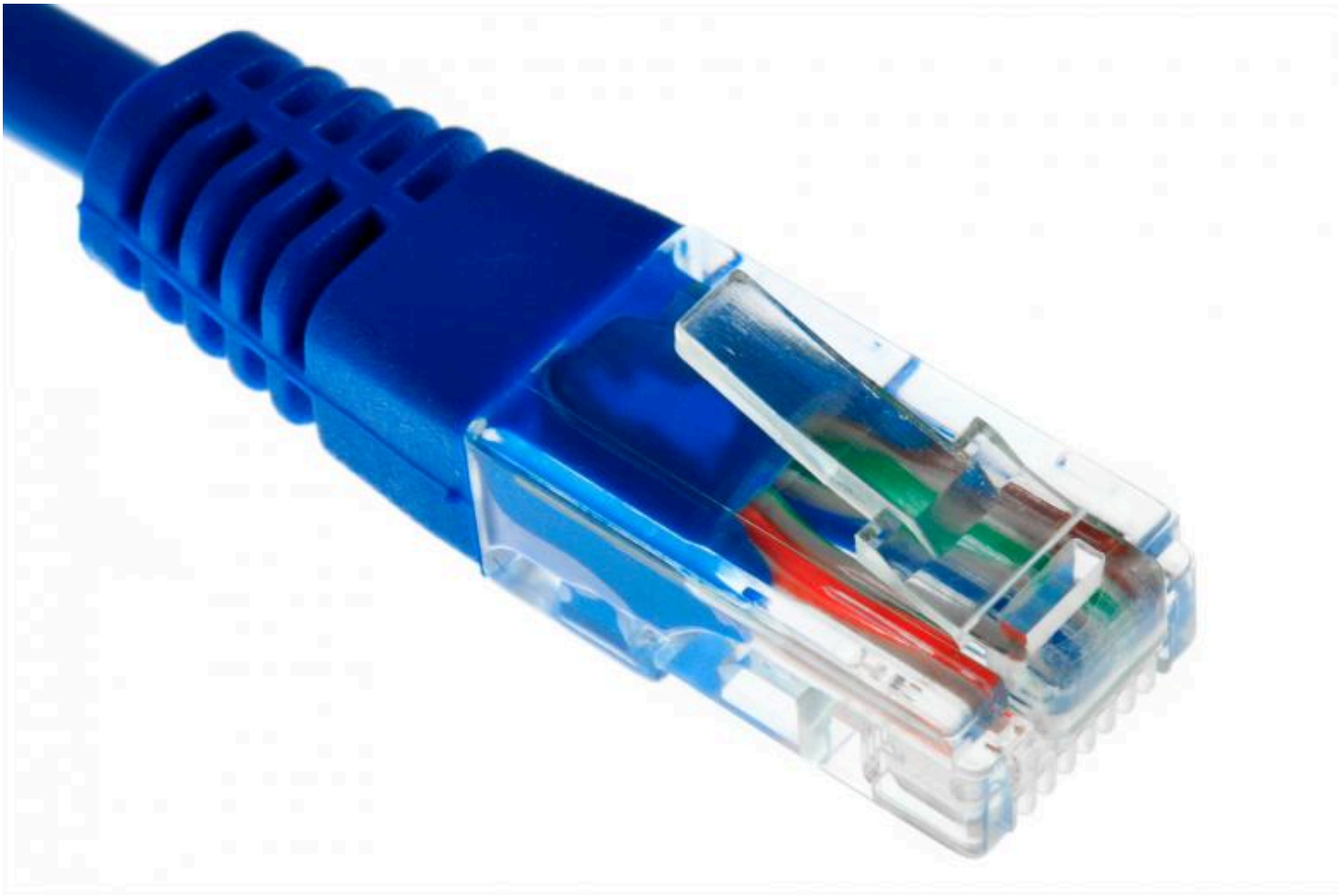
OSI Model



OSI Model – 1.Physical Layer (Layer 1)



The **Physical Layer** is the **first and lowest layer** of the OSI model.
It is responsible for transmitting **raw bits (0s and 1s)** over a physical medium.

It does **not** understand data, frames, or packets — only electrical, optical, or radio signals.

What Does Physical Layer Actually Do?

It deals with:

- Electrical signals
- Light pulses (fiber)
- Radio waves (Wi-Fi)
- Cables and connectors
- Voltage levels
- Bit timing
- Data transmission rate

Think of it as the **hardware transmission layer**.

Main Responsibilities

1.Bit Transmission

It converts:

0 → Low voltage
1 → High voltage

Or in fiber:

0 → No light
1 → Light pulse

It sends bits one by one over the medium.

2.Defines Physical Medium

It specifies:

- Type of cable (UTP, STP, Fiber)
- Connector type (RJ45)
- Maximum cable length
- Pin configuration
- Transmission speed (100 Mbps, 1 Gbps, etc.)

3.Bit Synchronization

Sender and receiver must agree on:

- When a bit starts
- When a bit ends
- Clock synchronization

Without timing → data becomes corrupted.

4.Transmission Modes

It defines communication type:

- Simplex → One way
- Half Duplex → Both ways, not at same time
- Full Duplex → Both ways simultaneously

5.Data Rate Control

Defines speed such as:

- 10 Mbps
- 100 Mbps
- 1 Gbps
- 10 Gbps

Devices Working at Physical Layer

- Hub
- Repeater
- Network cables
- Connectors
- Modem (partially)

Switch works mainly at Layer 2 but physically connects at Layer 1.

Real-World Example

When you plug an Ethernet cable:

1. Laptop converts data into binary.
2. Binary becomes electrical signals.
3. Signals travel through copper cable.
4. Router receives signals.
5. Converts signals back into binary.

All this is Physical Layer work.

What Physical Layer Does NOT Do

No error detection
No addressing
No data interpretation
No routing

It only moves bits.

Problems It Solves

Without Physical Layer:

- Devices couldn't transmit signals.
- No standardized voltage levels.
- No defined connectors.
- No communication medium.

It provides **standardized hardware communication rules**.

Technical Example

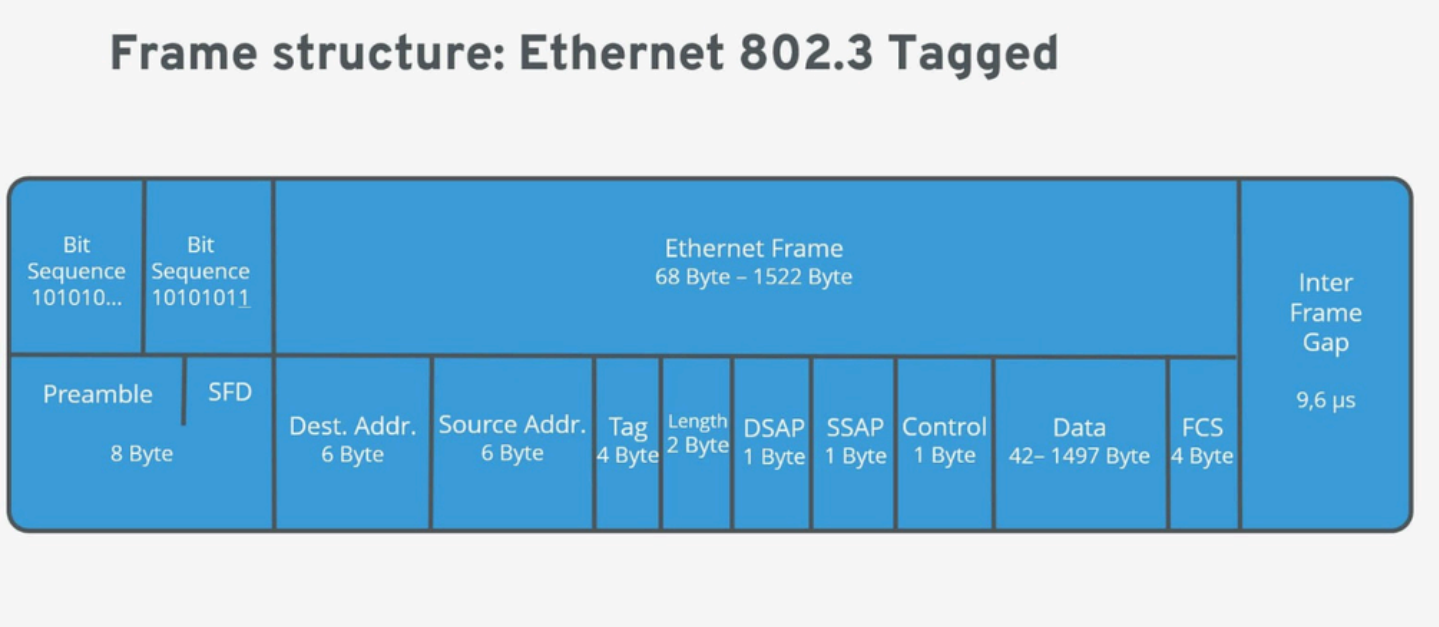
Example Ethernet standard:

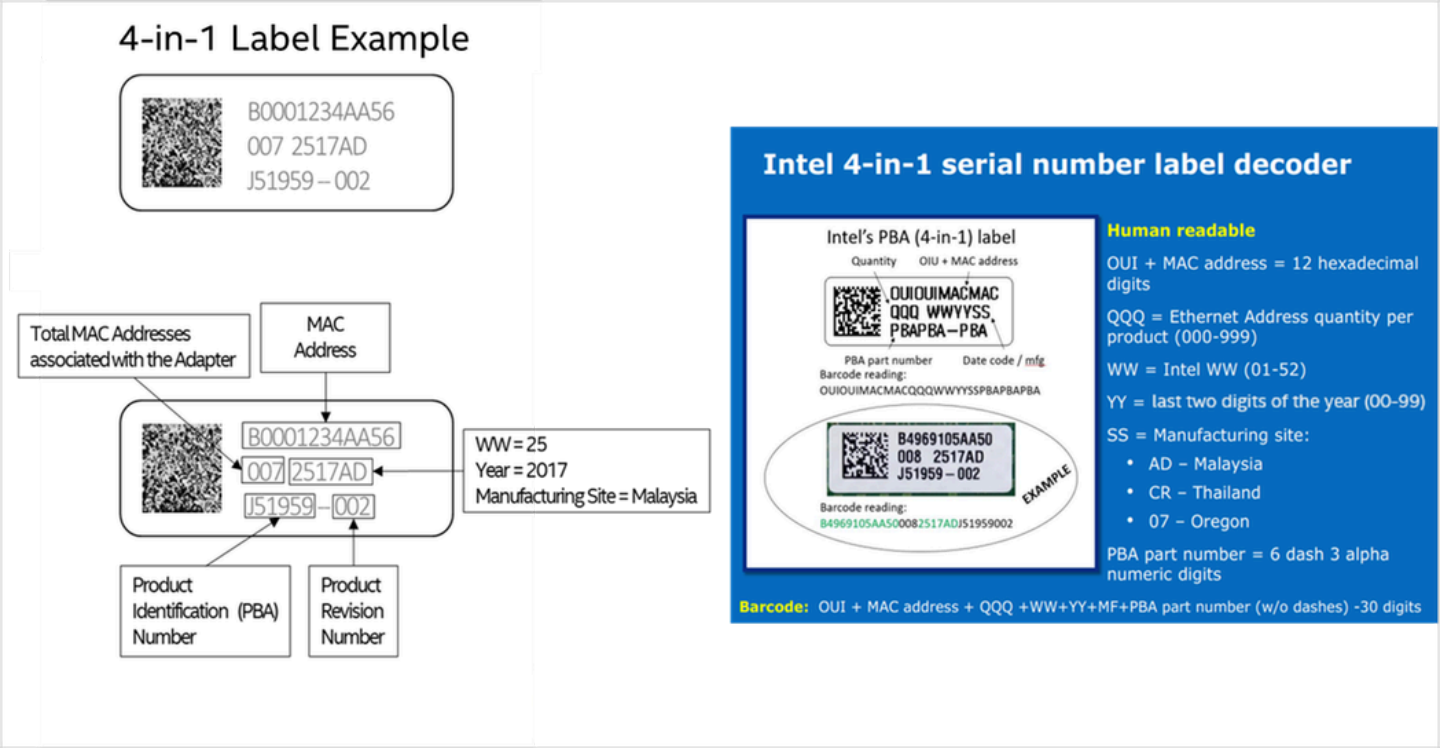
- 100BASE-TX
- Uses twisted pair cable
- 100 Mbps speed
- Max 100 meters

This is Physical Layer specification.

The Physical Layer is the first layer of the OSI model responsible for transmitting raw bits over a communication medium. It defines electrical signals, transmission rates, cable types, connectors, and physical topology.

OSI Model – 2.Data Link Layer (Layer 2)





The **Data Link Layer** is the second layer of the OSI model.
It works just above the Physical Layer.

If Physical Layer sends **raw bits**,
Data Link Layer makes those bits into **frames** and ensures they reach the correct device inside the same network.

What Does Data Link Layer Actually Do?

It is responsible for:

- Framing
- MAC addressing
- Error detection

- Flow control
- Local delivery (inside same network)

It works only within the **same LAN**.

Main Responsibilities

1.Framing

Physical layer sends only bits:

010101010101010

Data Link Layer groups bits into a **frame**:

| Header | Data | Trailer |

This structure is called an **Ethernet Frame**.

2.MAC Addressing

Every device has a **MAC address** (Physical address).

Example:

00:1A:2B:3C:4D:5E

When sending data inside LAN:

- Source MAC
- Destination MAC

are added in the frame.

This ensures the frame reaches the correct device.

3.Error Detection

Data Link Layer checks:

- Did data change during transmission?
- Is frame corrupted?

It uses:

- CRC (Cyclic Redundancy Check)

If error found → frame is discarded.

4.Flow Control

Prevents a fast sender from overwhelming a slow receiver.

It controls speed of transmission.

5.Media Access Control

If multiple devices share same cable:

- Who can send first?
- Who should wait?

Protocols like:

- CSMA/CD (Ethernet)
- CSMA/CA (WiFi)

handle this.

Devices Working at Data Link Layer

- Switch
- Bridge
- Network Interface Card (NIC)

Switch uses MAC address table to forward frames.

Real-Life Example

Inside a classroom:

- Students = devices
- Roll number = MAC address
- Teacher sending message = frame transmission

Teacher calls specific roll number →
Only that student responds.

That is Data Link Layer working.

What Data Link Layer Does NOT Do

No IP addressing
No routing between networks

No logical addressing
No end-to-end delivery

It works only inside local network.

Problems It Solves

Without Data Link Layer:

- No device identification inside LAN
- No frame structure
- No local error checking
- No proper switching

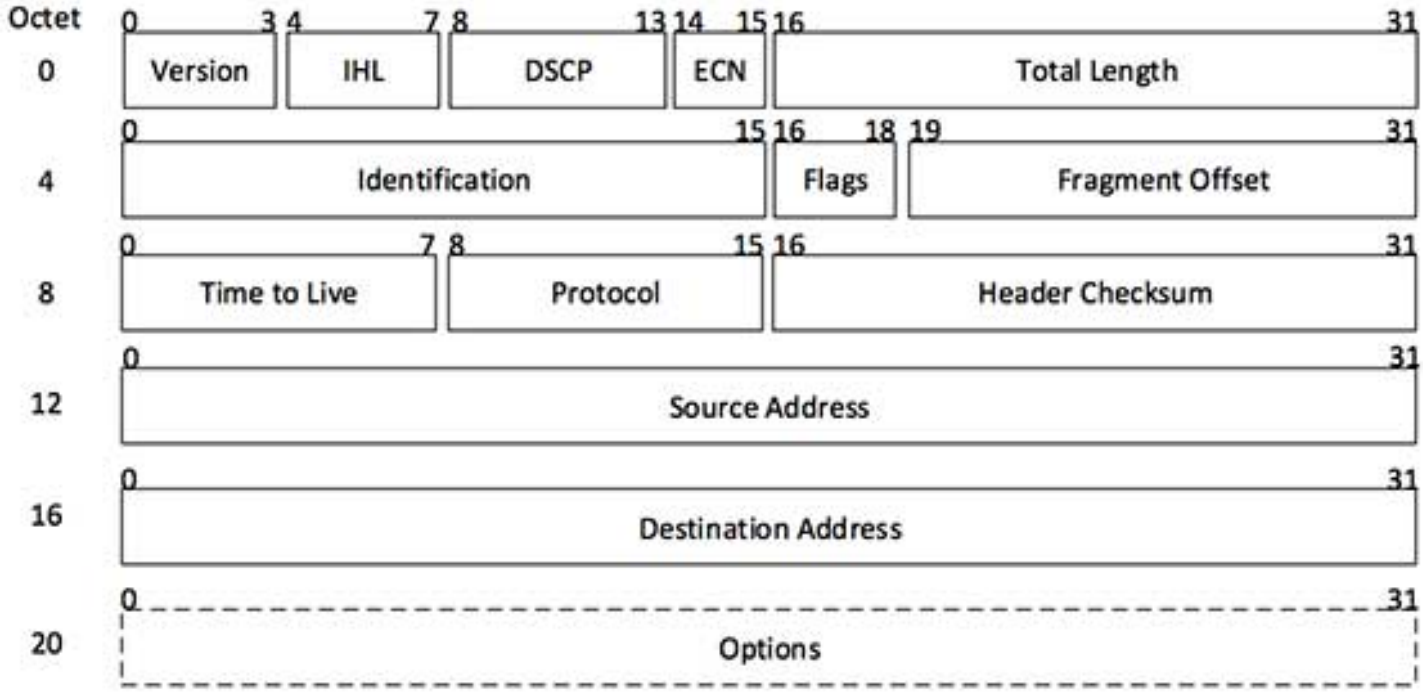
Physical layer alone is not enough.

Physical vs Data Link

Physical Layer	Data Link Layer
Sends raw bits	Creates frames
Deals with cables	Deals with MAC
No error detection	Detects errors
No addressing	Uses MAC address

The Data Link Layer is responsible for framing, MAC addressing, error detection, and reliable data transfer within the same local network.

OSI Model – 3.Network Layer (Layer 3)



[Image: IP Header]

The **Network Layer** is the third layer of the OSI model.

If:

- Physical Layer → sends bits
- Data Link Layer → sends frames inside LAN
- Network Layer → sends packets between different networks

It is responsible for **logical addressing and routing**.

What Does Network Layer Actually Do?

It handles:

- IP addressing
- Routing
- Packet forwarding
- Path selection
- Fragmentation

It allows communication between different networks.

Main Responsibilities

1.Logical Addressing (IP Address)

Every device gets an **IP address**.

Example:

192.168.1.10

Unlike MAC address (physical),
IP address is **logical and changeable**.

It identifies:

- Network
- Device

2.Packet Creation

Data from upper layer becomes:

| IP Header | Data |

This structure is called an **IP Packet**.

Header contains:

- Source IP
- Destination IP
- TTL
- Protocol

3.Routing (Most Important)

When sending data to another network:

Device checks:

- Is destination in same network?

If not:

- Send packet to router

Router decides best path using routing table.

Example path:

Laptop → Router → ISP → Internet → Destination

4.Path Selection

Routers choose shortest or best path using routing protocols like:

- RIP

- OSPF
- BGP

This is fully Network Layer responsibility.

5.Fragmentation

If packet size is larger than network limit:

- It breaks packet into smaller pieces.
- Destination reassembles them.

Devices Working at Network Layer

- Router
- Layer 3 Switch

Router reads IP address and forwards packet.

Real-Life Example

Think of postal system:

- MAC address → House number
- IP address → City name

If sending inside same city → local delivery
If sending to another city → needs main post office (router)

Router = Post office that decides route.

What Network Layer Does NOT Do

No error correction
No frame creation
No bit transmission
No session management

It only handles logical addressing and routing.

Problem It Solves

Without Network Layer:

- No communication between different networks
- No internet

- No routing
- No global addressing system

LAN would exist, but internet would not.

Data Flow Example (Step-by-Step)

Suppose:

Your IP: 192.168.1.10
Google IP: 142.250.x.x

Steps:

1. Your device creates IP packet.
2. Checks subnet mask.
3. Destination not in same network.
4. Sends packet to router.
5. Router forwards based on routing table.
6. Packet reaches Google.

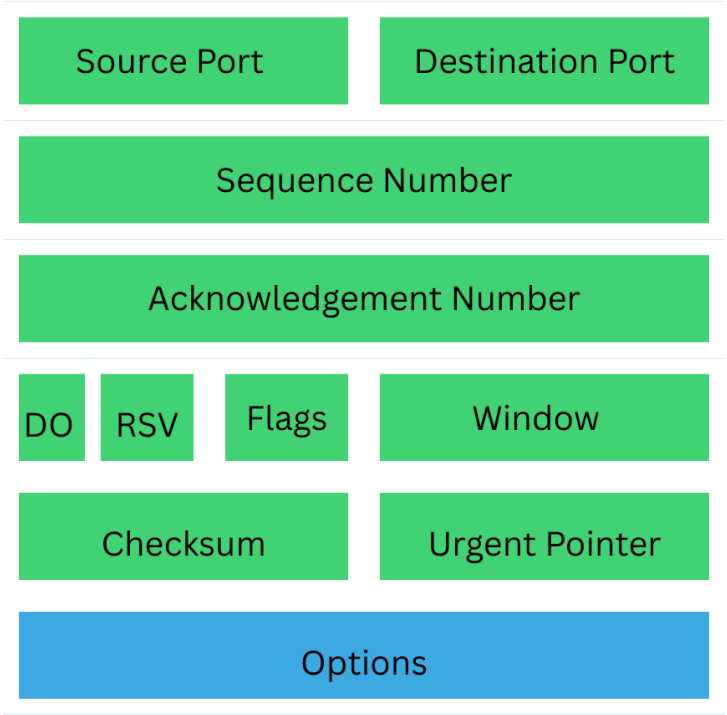
All this is Network Layer work.

Data Link vs Network Layer

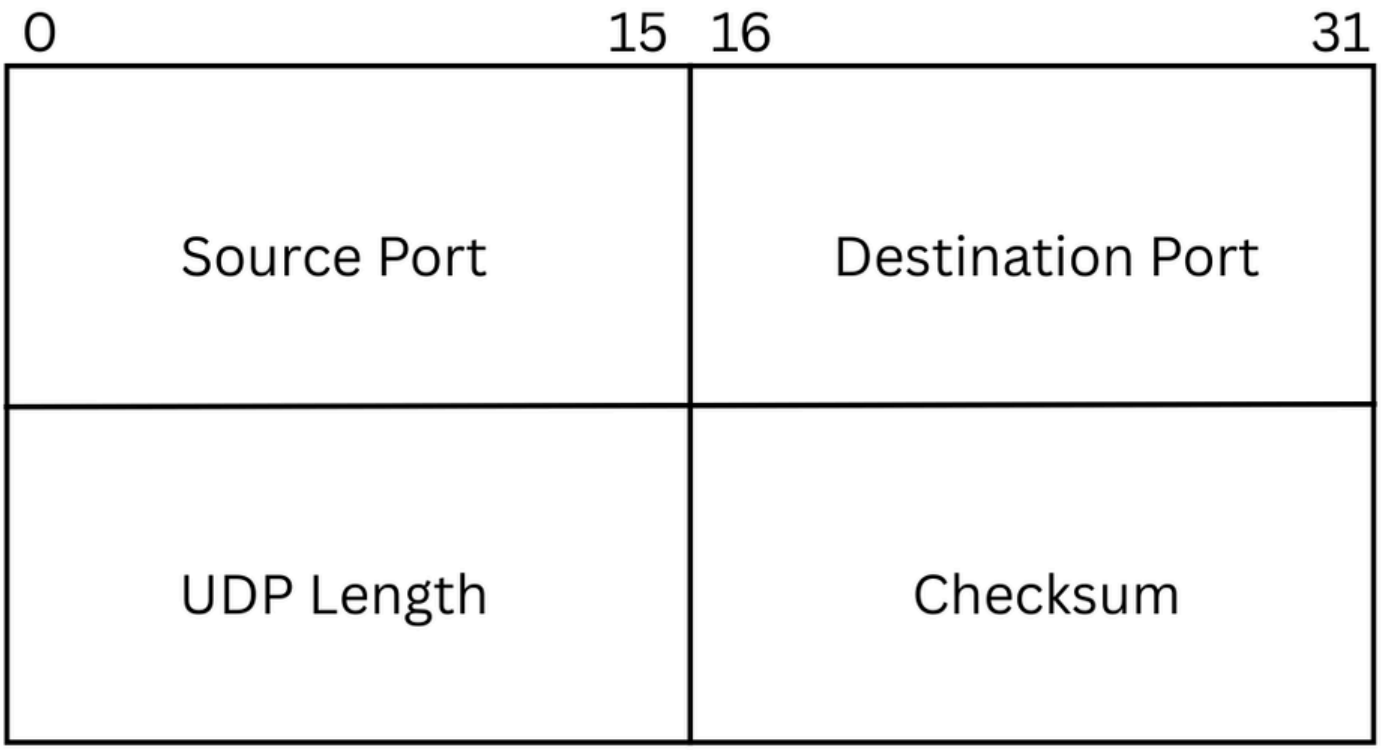
Data Link Layer	Network Layer
Uses MAC address	Uses IP address
Works inside LAN	Works between networks
Frame delivery	Packet delivery
Switch works here	Router works here

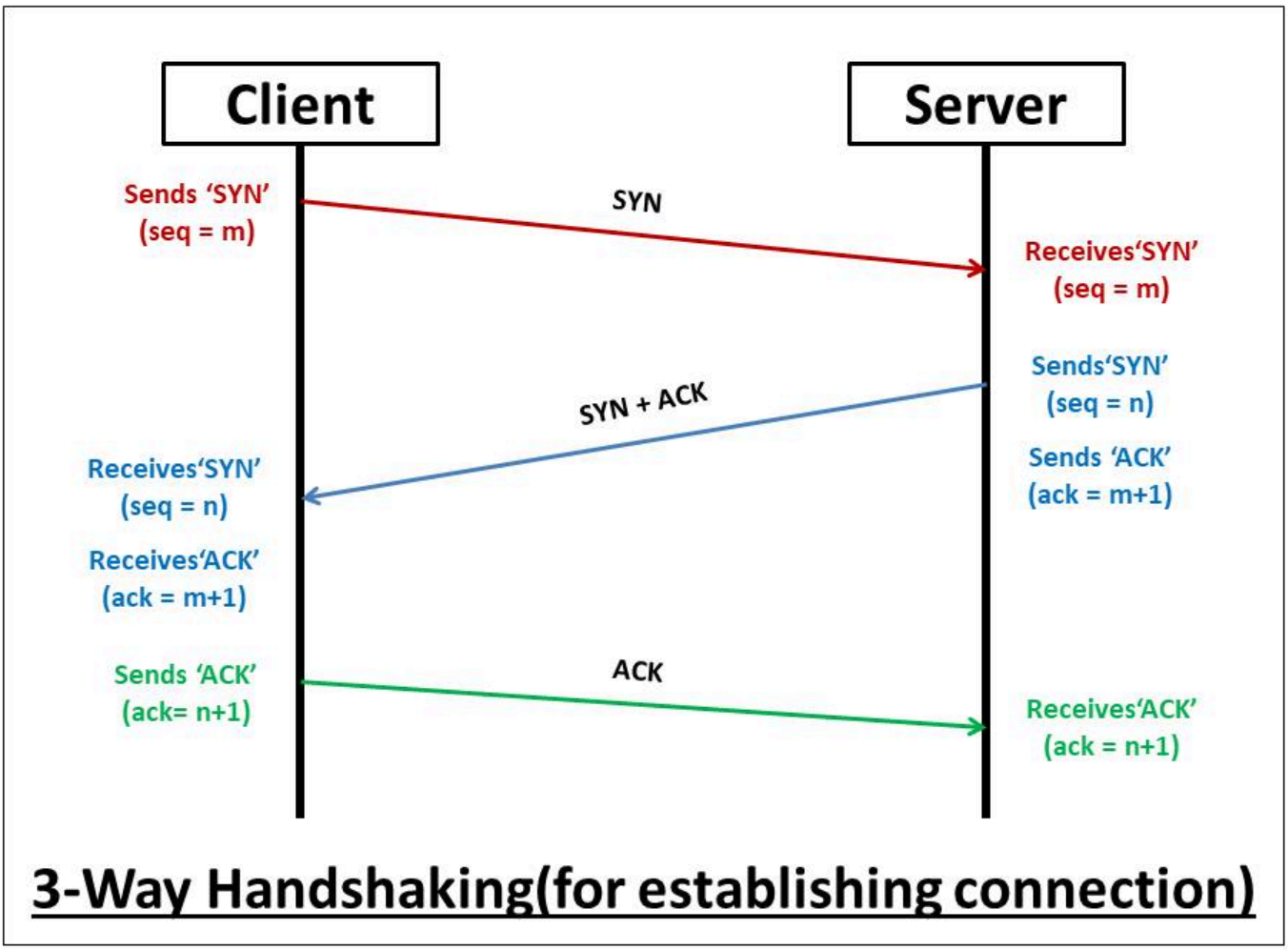
The Network Layer is responsible for logical addressing, routing, and forwarding packets between different networks. It ensures data travels from source to destination across multiple interconnected networks.

OSI Model – 4.Transport Layer (Layer 4)

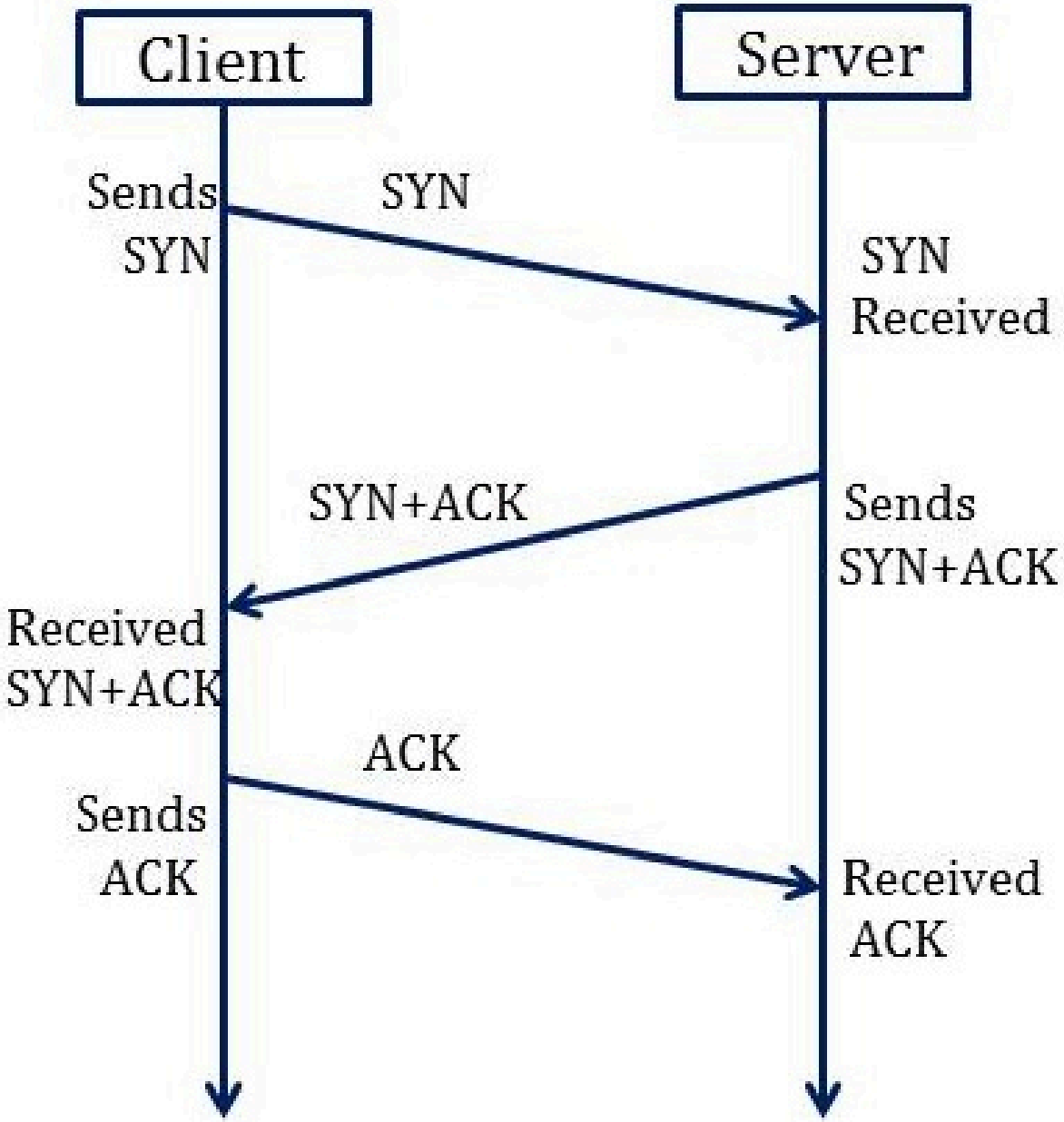


UDP Header Format





3-Way Handshaking(for establishing connection)



The **Transport Layer** is the fourth layer of the OSI model.

If:

- Physical → sends bits
- Data Link → sends frames
- Network → sends packets
- Transport → ensures reliable end-to-end delivery

This layer works between **source application and destination application**.

What Does Transport Layer Actually Do?

It handles:

- End-to-end communication
- Reliability
- Port numbers
- Error recovery
- Flow control
- Congestion control
- Segmentation

It makes sure data reaches correctly and completely.

Main Responsibilities

1.Segmentation

Large data is broken into smaller parts called **segments**.

Example:

Big file → Segment 1 → Segment 2 → Segment 3

Each segment is sent separately.

At destination → segments are reassembled.

2.Port Numbers

IP address identifies device.
Port number identifies application.

Example:

IP: 192.168.1.10
Port: 80 (HTTP)
Port: 443 (HTTPS)
Port: 21 (FTP)

So Transport Layer ensures:

- Data goes to correct app
- Not just correct device

3.Reliability (TCP)

Transport Layer ensures:

- Data arrives
- No duplication

- No missing segments
- Correct order

If segment lost → resend.

This is done using **TCP**.

4.Fast Communication (UDP)

Sometimes speed is more important than reliability.

Example:

- Live streaming
- Online gaming
- Video calls

For this, **UDP** is used.

UDP:

- No guarantee
- No retransmission
- Faster

Two Main Protocols

TCP (Transmission Control Protocol)

Reliable
Ordered
Error checking
Three-way handshake
Slower but safe

Used for:

- Web browsing
- Email
- File transfer

UDP (User Datagram Protocol)

Fast
No connection setup
No guarantee
Lightweight

Used for:

- Streaming
- Gaming
- DNS

TCP Three-Way Handshake (Connection Setup)

Before sending data:

1. SYN → Client says “Can we connect?”
2. SYN-ACK → Server says “Yes”
3. ACK → Client confirms

Connection established.

Flow Control

Prevents sender from overwhelming receiver.

Uses:

- Window size

Receiver tells:

“Send only this much data at a time.”

Congestion Control

If network is busy:

- TCP reduces speed
- Avoids network collapse

Very important for internet stability.

Real-Life Example

Think of courier service:

IP Address → City
Port → Specific office

TCP → Registered post (safe, signed, tracked)
UDP → Normal postcard (fast, no guarantee)

What Transport Layer Does NOT Do

No routing
No IP addressing
No cable transmission
No MAC addressing

It works above Network Layer.

Problem It Solves

Without Transport Layer:

- Data could arrive incomplete
- Order may change
- No app-level delivery
- No reliability
- No speed control

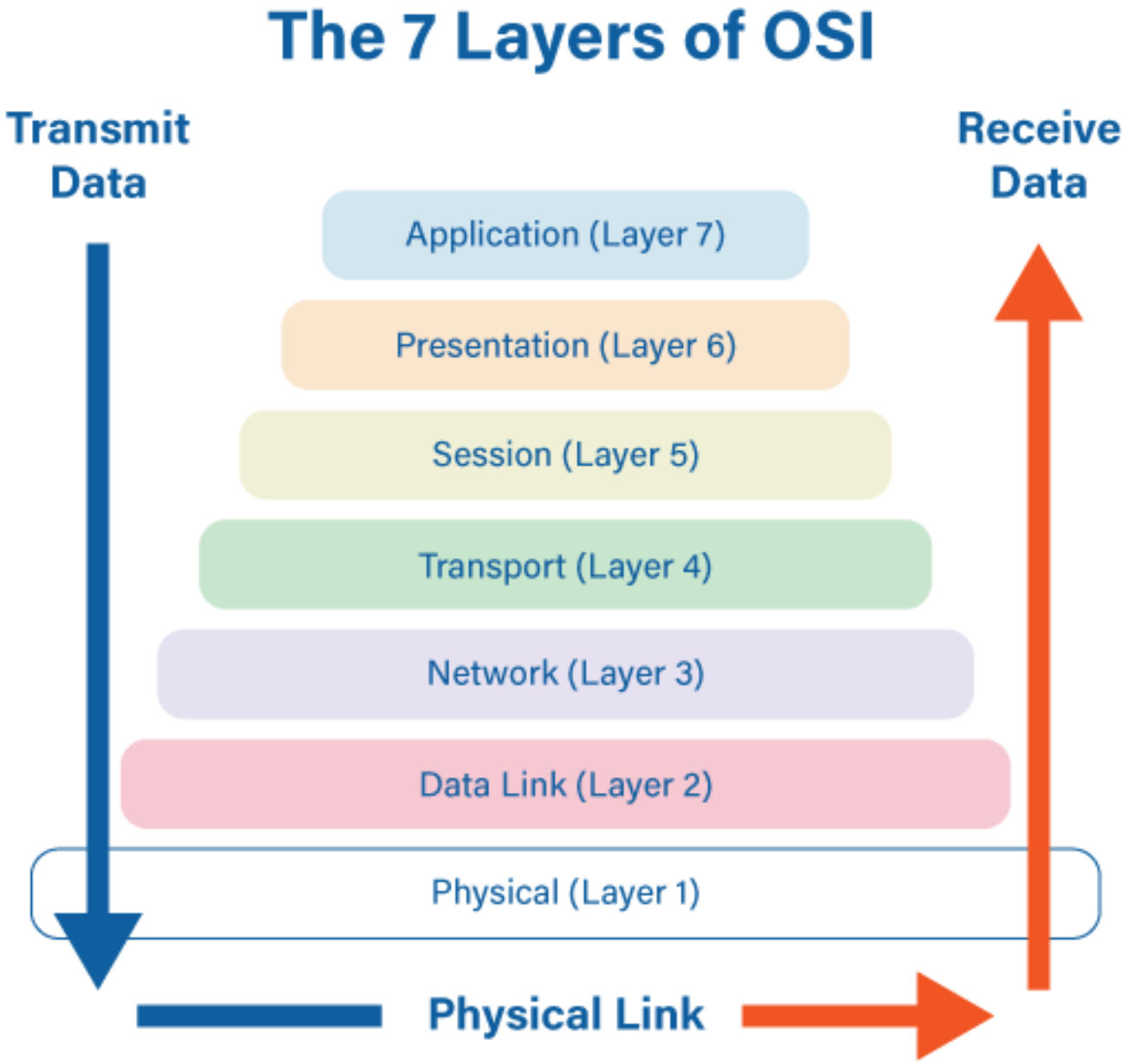
Internet would be unstable.

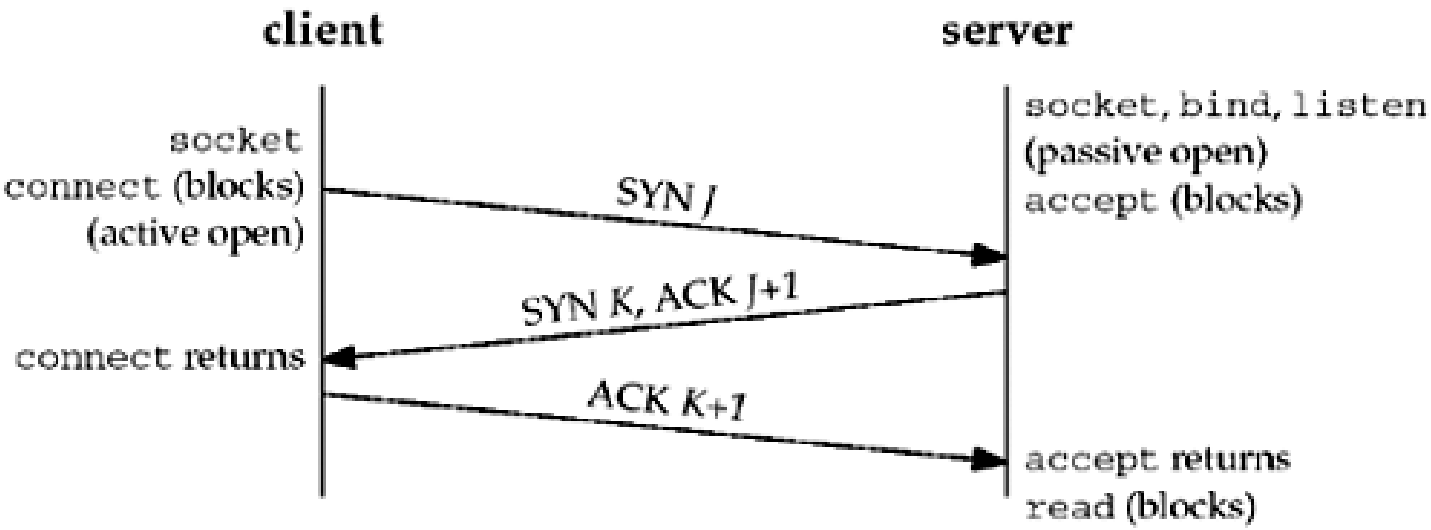
Network vs Transport Layer

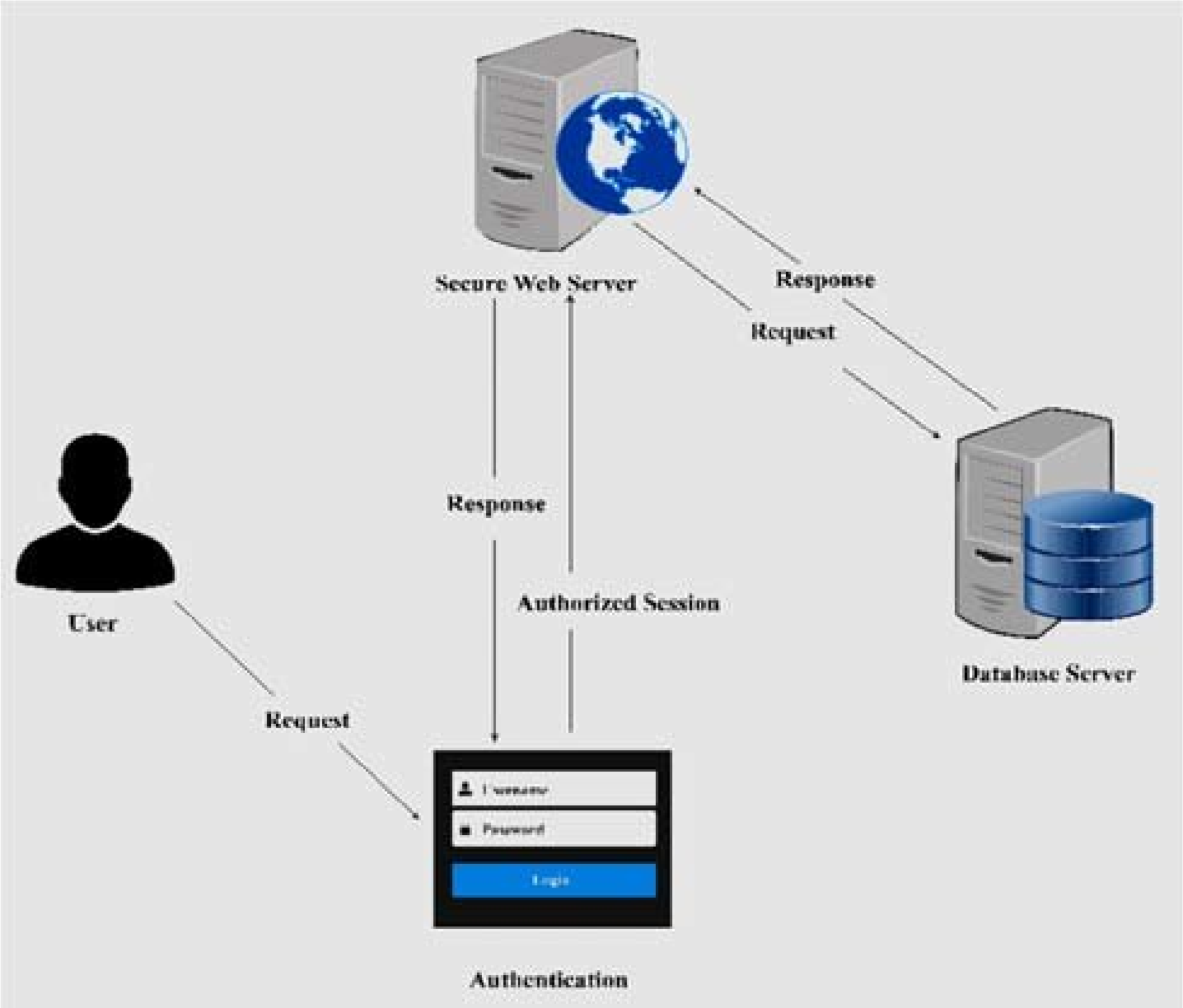
Network Layer	Transport Layer
Device to device	Application to application
Uses IP	Uses Port number
Router works here	End devices handle this
Packet forwarding	Reliability & control

The Transport Layer ensures reliable end-to-end communication between applications. It handles segmentation, port addressing, flow control, congestion control, and error recovery using protocols like TCP and UDP.

OSI Model – 5.Session Layer (Layer 5)


The 7 Layers of OSI
Transmit Data
Receive Data
Application (Layer 7)
Presentation (Layer 6)
Session (Layer 5)
Transport (Layer 4)
Network (Layer 3)
Data Link (Layer 2)
Physical (Layer 1)
Physical Link


client
server
socket
connect (blocks)
(active open)
socket, bind, listen
(passive open)
accept (blocks)
SYN J
SYN K, ACK J+1
connect returns
ACK K+1
accept returns
read (blocks)



The **Session Layer** is the fifth layer of the OSI model.

If:

- Network → finds the path
- Transport → delivers data reliably
- Session → manages the conversation

It controls the **start, management, and end of communication sessions** between two devices.

What Does Session Layer Actually Do?

It handles:

- Session establishment
- Session maintenance
- Session termination
- Dialog control
- Synchronization (checkpoints)

It keeps communication organized.

What Is a Session?

A session is a **conversation between two applications**.

Example:

- You log into a website
- You start a video meeting
- You access a remote server

That connection is called a session.

Main Responsibilities

1.Session Establishment

Before communication:

- It sets up the session.
- Decides how communication will happen.

Example:

Client → “Start session”
Server → “Session accepted”

2.Session Maintenance

While communication is happening:

- Keeps track of session
- Ensures connection stays active

If connection temporarily drops:

- Session Layer helps resume.

3.Session Termination

After work is done:

- Closes session properly
- Frees resources

Example:

Logout → Session closed

4.Dialog Control

Controls who can send data and when.

Two types:

- Half-duplex (one at a time)
- Full-duplex (both can talk simultaneously)

5.Synchronization (Checkpoints)

If large data transfer happens:

- Session layer sets checkpoints.
- If failure occurs, resume from last checkpoint.

Example:

Downloading 5GB file:

- Failure at 4GB
- Resume from 4GB (not from 0)

Real-Life Example

Think of a phone call:

- Dialing → Session establishment
- Talking → Session maintenance
- Hanging up → Session termination

Without session control:

- Calls could mix
- Conversations would not end properly

Example in Real World Applications

- Online banking login
- Remote desktop connection
- Video conferencing
- File download resume

What Session Layer Does NOT Do

No routing
No error correction
No IP addressing
No data formatting

It only manages communication sessions.

Problem It Solves

Without Session Layer:

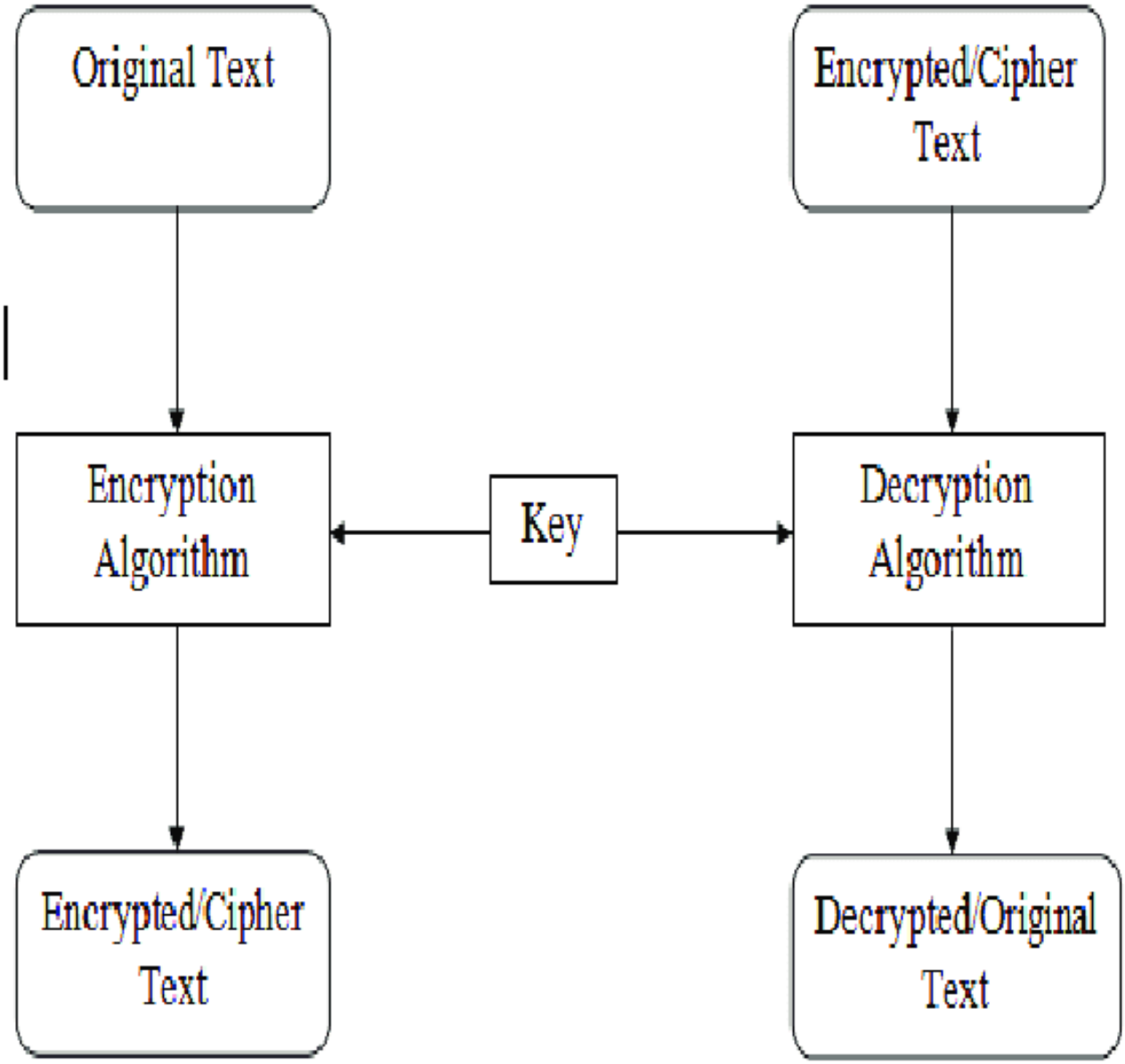
- No proper login/logout handling
- Interrupted transfers would restart from beginning
- No control over who talks when
- No structured communication

Transport vs Session Layer

Transport Layer	Session Layer
Reliable delivery	Manages conversation
Uses TCP/UDP	Controls session logic
Handles ports	Handles login/logout
Focus on data	Focus on communication state

The Session Layer establishes, manages, and terminates communication sessions between applications. It provides dialog control and synchronization to ensure organized communication.

OSI Model – 6.Presentation Layer (Layer 6)



The **Presentation Layer** is the sixth layer of the OSI model.

If:

- Session → manages conversation
- Presentation → formats the data
- Application → interacts with user

This layer makes sure data is in a **readable and usable format**.

What Does Presentation Layer Actually Do?

It handles:

- Data translation
- Data encryption
- Data decryption

- Data compression
- Data decompression

It works like a **translator and formatter**.

Main Responsibilities

1.Data Translation

Different systems use different formats.

Presentation layer converts:

- ASCII ↔ Unicode
- JPEG ↔ Bitmap
- Different data formats

Example:

Sender format → Common format → Receiver format

So both systems understand each other.

2.Encryption

Before sending data:

- It converts plain text into cipher text.
- Protects data from hackers.

Example:

"Hello" → Encrypted code

Used in:

- Online banking
- HTTPS websites
- Secure login

3.Decryption

At receiver side:

- Converts encrypted data back to original form.

Without this, data would look unreadable.

4.Compression

Reduces size of data before sending.

Benefits:

- Faster transmission
- Less bandwidth usage

Example:

ZIP files, image compression.

5.Decompression

Receiver expands compressed data back to original size.

Real-Life Example

Think of a translator in a meeting:

- One person speaks Hindi
- Other speaks English

Translator converts language so both understand.

Presentation Layer = Data Translator.

Another example:

Think of encryption like locking a box before sending it.

Real World Applications

- HTTPS (SSL/TLS encryption)
- Image formats (JPEG, PNG)
- Audio formats (MP3)
- Video formats (MP4)
- File compression (ZIP)

What Presentation Layer Does NOT Do

No routing
No port management
No session control
No bit transmission

It only handles data format and security.

Problem It Solves

Without Presentation Layer:

- Systems using different formats could not communicate
- No secure data transfer
- No encryption
- No compression

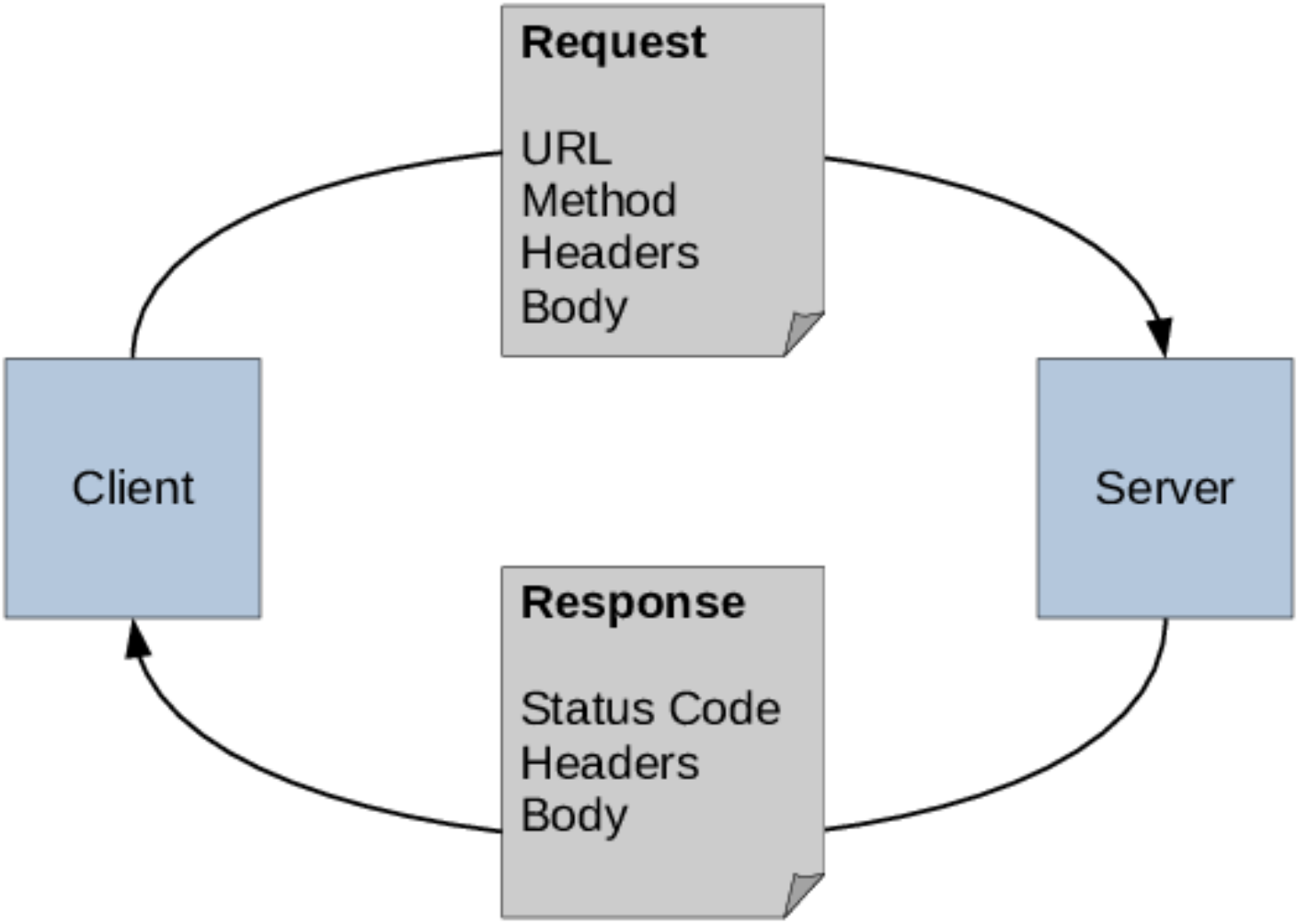
Internet would be unsafe and inefficient.

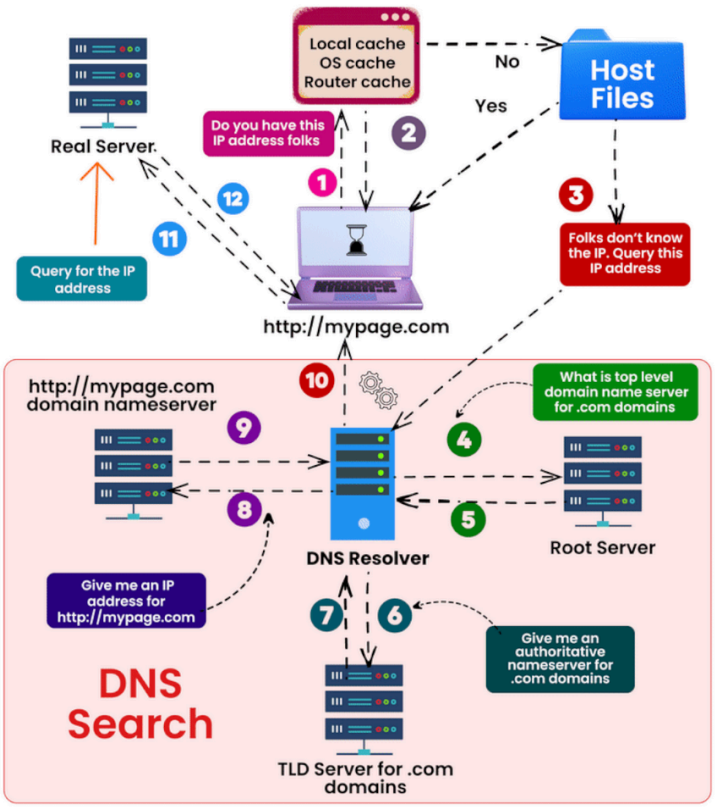
Session vs Presentation Layer

Session Layer	Presentation Layer
Manages conversation	Formats data
Login/logout control	Encryption
Session checkpoints	Compression
Dialog management	Data translation

The Presentation Layer is responsible for data translation, encryption, decryption, and compression. It ensures that data sent by the application layer of one system is readable and secure for the application layer of another system.

OSI Model – 7.Application Layer (Layer 7)





The **Application Layer** is the top (7th) layer of the OSI model.

If:

- Presentation → formats data
- Application → interacts with user

This layer provides **network services directly to end users and applications**.

It is the closest layer to the user.

What Does Application Layer Actually Do?

It provides:

- Web access
- Email services
- File transfer
- Name resolution
- Remote login

It allows software to communicate over the network.

Important Note

Application Layer does NOT mean:

- Microsoft Word
- Chrome browser
- WhatsApp

Those are applications.

Application Layer provides the **protocols** that these applications use.

Main Responsibilities

1.Web Communication (HTTP / HTTPS)

When you open a website:

- Browser sends HTTP request.
- Server sends response.

Example:

Client → GET /index.html
Server → HTML page

2.Email Services

Protocols:

- SMTP → Send email
- POP3 → Receive email
- IMAP → Manage email

3.File Transfer (FTP)

Used to upload/download files between systems.

Example:

Client uploads file → Server stores file

4.DNS (Domain Name System)

Converts:

google.com → 142.250.x.x

Without DNS, we would need to remember IP addresses.

5.Remote Access

Protocols like:

- Telnet
- SSH

Allow remote login to another computer.

Common Application Layer Protocols

- HTTP / HTTPS
- FTP
- SMTP
- DNS
- POP3
- IMAP
- Telnet
- SSH

Real-Life Example

Think of Application Layer like a receptionist:

You tell receptionist what you need:

- “Open website”
- “Send email”
- “Download file”

Receptionist connects you to the correct service.

What Application Layer Does NOT Do

No routing
No encryption (handled by Presentation layer)
No reliability control (Transport layer)
No session management

It only provides services to user applications.

Problem It Solves

Without Application Layer:

- No web browsing
- No email
- No file transfer
- No domain name system

Users would not be able to use the network easily.

Full OSI Layers Quick Summary

Layer	Function
7 Application	User services
6 Presentation	Format & encryption
5 Session	Manage session
4 Transport	Reliable delivery
3 Network	Routing
2 Data Link	Frame delivery
1 Physical	Bits transmission

The Application Layer provides network services directly to end-user applications. It enables web browsing, email, file transfer, and other network-based services using protocols like HTTP, FTP, SMTP, and DNS.